DATA SHEET :

MCR SECOND STAGE COMPRESSOR
H4K-3

REV. NO	APPROVED BY	DATE
		1/5/97

FINAL APPROVAL BY PKP

CLIENT : PERUSAHAAN PERTAMBANGAN MINYAK DAN GAS BUMI NEGARA (PERTAMINA)/ PROYEK KILANG PENGOLAHAN (PKP)

JOB TITLE : BONTANG LNG EXPANSION PROJECT TRAIN H

PROJECT NAME : PGH PROJECT

LOCATION : BONTANG, EAST KALIMANTAN, INDONESIA

FOR RECORD

REV	DATE	PAGE	DESCRIPTION	PREP'D	CHK'D	APP'D
0	19 Dec.1996	ALL	Issued for Approval and Inquiry	ADN SDP	AS	BS JGB
1	29 Jan. 1997	ALL	For PKP review during Bid Clarification Meeting	ADN SDP	AS SWK	
2	6 Feb. 1997	ALL	Issued for Approval based on PKP comments at Bid Clarification Meeting	ADN IRA	AS SWK	BS JGB
3	24 Mar. 1997	ALL	Issued for PKP Approval revised based on PKP comments HX-PJIP-0049	ADN	AS SWK	BS JGB
4	16 Apr. 1997	10	Approved per HX-PJIP-0112/0165 and issued for Purchase Order	ADN		

IKPT / KELLOGG

CENTRIFUGAL COMPRESSOR DATA SHEET SI (METRIC) UNITS

JOB NO. 95038 / 6413 ITEM NO. H4K-3
P.O. NO. DATE 16 April 1997
REQUISITION NO. PGH-MR-K-003
DOC. NO. DS-H04-K-003

APPLICABLE TO : PROPOSAL ● PURCHASE AS BUILT DATE 16 April 1997 REVISION 4
FOR BONTANG LNG EXPANSION PROJECT, TRAIN-H UNIT PLANT 4 - REFRIGERATION
SITE BONTANG, EAST KALIMANTAN, INDONESIA SERIAL NO.
SERVICE MR FIRST STAGE COMPRESSOR NO. REQUIRED ONE
MANUFACTURER DRESSER - RAND MODEL 656 B7-6 DRIVER STEAM TURBINE (H4KT-3)
NOTE : INFORMATION TO BE COMPLETED : BY PURCHASER BY MANUFACTURER.

OPERATING CONDITIONS

(ALL DATA ON PER UNIT BASIS)

	NORMAL & RATED			ALT. OPERATION		
	CASE 1			CASE 2	CASE 3	Recycle (2) by Vendor
◉ GAS HANDLED (Also See Page 3) nm3/h (1.013 bar & 0°C Dry)	712266			705833	703421	
◉ Weight Flow, kg/h (Dry)	759,018 x 1.025			751,221	748,735	
INLET CONDITIONS						
◉ Pressure, (Kg/cm2) A	15.44			15.29	15.22	
◉ Temperature, ° C	32.70			32.70	32.70	
○ Mix Temperature °C						
◉ Molecular Weight, M	25.075			25.044	25.046	
◉ Cp / Cv (K1)	1.207			1.206	1.206	
☒ Compressibility (Z1)	.938			.939	.939	
☒ Inlet Volume, m3/h (Dry)	48897			48973	49042	
DISCHARGE CONDITIONS						
☒ Pressure, (Kg/cm2) A	54.84			54.84	54.84	
☒ Temperature, ° C (to be verified by vendor)	126.9			127.8	127.8	
☒ Cp / Cv (K2) AVG	1.207			1.206	1.206	
☒ Compressibility (Z2) AVG	.927			.928	.928	
☒ KW Required (All Losses Included) ± 4 %	36681			36656	36659	
☒ Speed, rpm	4445			4461	4467	
☐ Estimated Surge, m3/h (at Speed Above)						
☒ Polytropic Head, M-kg/kg	13945			14094	14150	
☒ Polytropic Efficiency, %	80.8			80.9	80.9	
◉ Guarantee Point See note 1 , page 14 of 16	YES					
☒ Performance Curve No.	see pages	3, 4, 5, 6, 7, 8 of 16				

PROCESS CONTROL By HCJV

Method : ○ Suction Throttling From ____ (bar) To ____ (bar) ○ Variable Inlet Guide Vanes (3.4.2.4) ◉ Speed Variation From 80 % To 105 % ○ Discharge Blowoff To ____ ◉ Cooled Bypass From Discharge To Suction

Signal ◉ Source (3.4.2.1) MANUAL CONTROLLER (Electronic Signal/hydraulically Actuated) FOR SPEED CONTROL

Type ○ Electronic ○ Pneumatic ◉ Other ANTI-SURGE CONTROLLER (Electronic Signal / Pneumatically Actuated) FOR COOLED BY PASS

Range 4-20 mA 0.2-1.0 (kg/cm2G)

Antisurge Bypass : ○ Manual ◉ Automatic ○ None

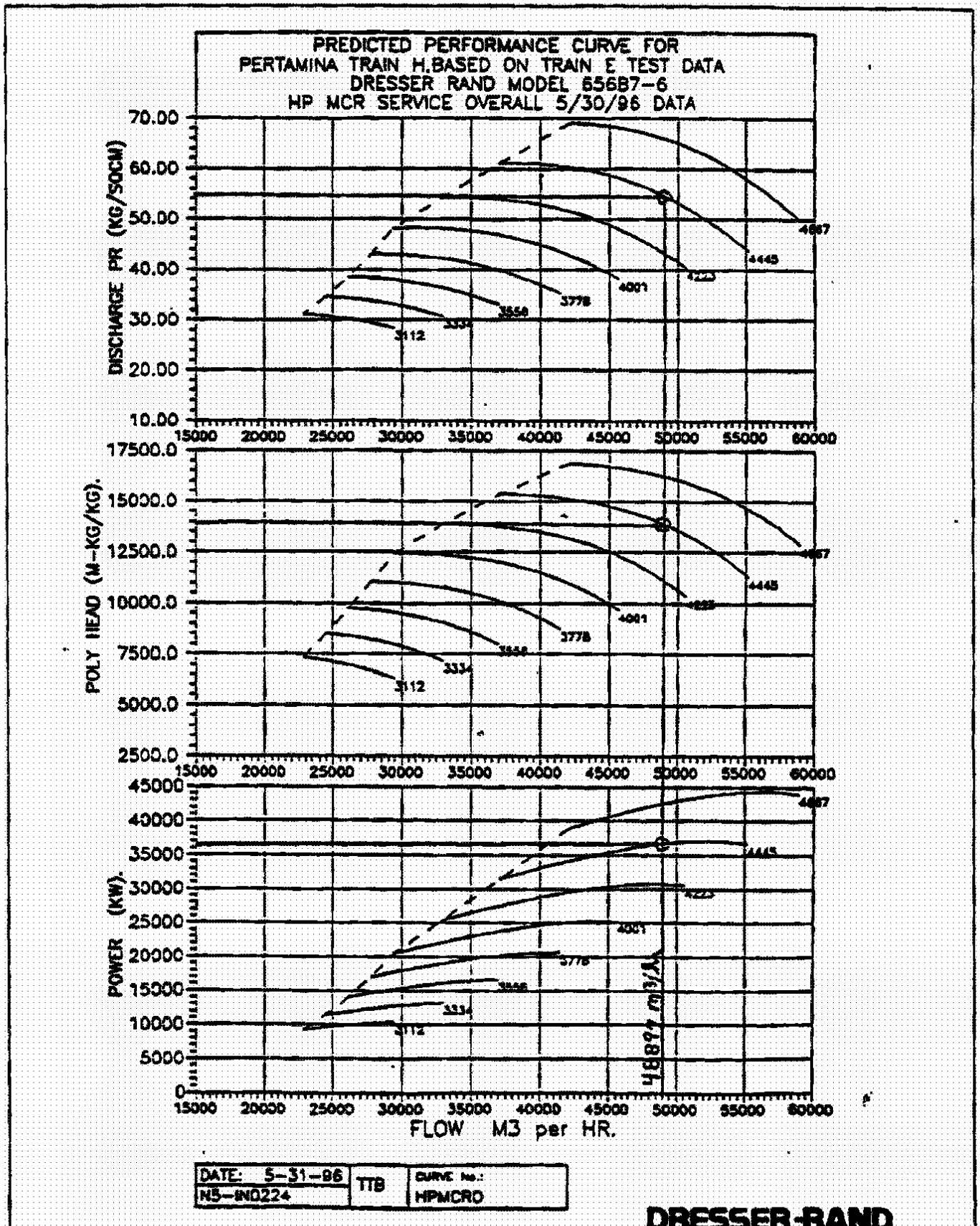
REMARKS

(1) VENDOR SHALL CONFIRM INLET AND DISCHARGE GAS PROPERTIES.
(2) VENDOR SHALL CONFIRM COMPRESSOR PERFORMANCE DURING FULL RECYCLE. GAS TEMPERATURE UPSTREAM OF ANTI-SURGE CONTROL VALVE IS 32.4°C AT 53.65 KG/CM2 (ABS) FOR CASES 1, 2 AND 3 .
RECYCLE GAS TO BE EXPANDED TO SUCTION PRESSURE FOR EACH RESPECTIVE CASE.
(3) CASE 1 : MINIMUM BUTANE RECYCLE, CASE 2 : MAXIMUM BUTANE RECYCLE, CASE 3 : NO LPG PRODUCTION
(4) VENDOR TO COMPLETE THE DATA IN THE BLANK CELLS

IKPT KELLOGG

CENTRIFUGAL COMPRESSOR
DATA SHEET
SI (METRIC) UNITS

JOB NO. 95038/6413 ITEM NO. H4K-3
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REQUISITION NO. PGH-MR-K-003
DOC. NO. DS-H04-K-003



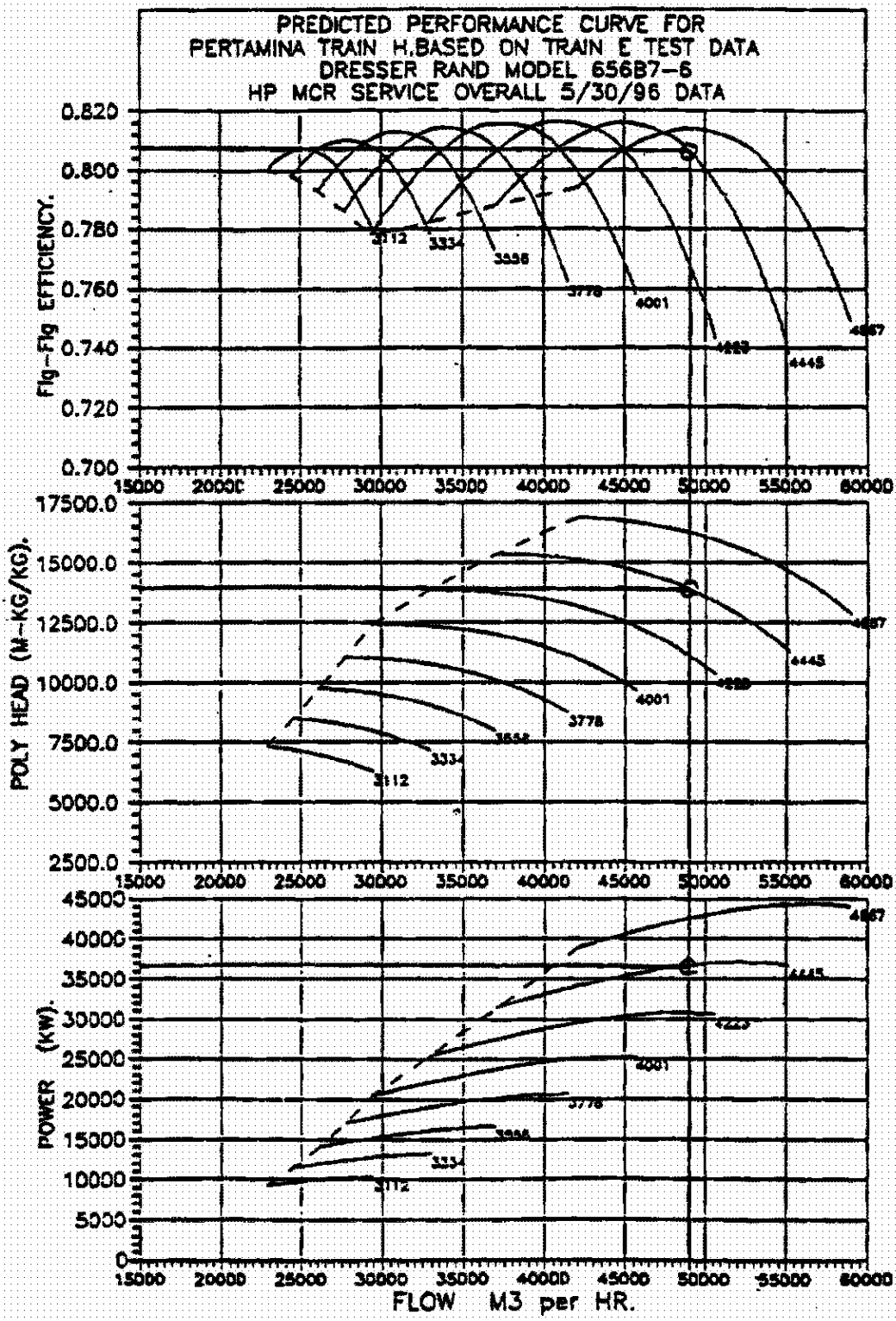
DATE: 5-31-96	TTB	CURVE No.:
N5-IN0224		HPMCRD

DRESSER-RAND

IKPT KELLOGG

CENTRIFUGAL COMPRESSOR
DATA SHEET
SI (METRIC) UNITS

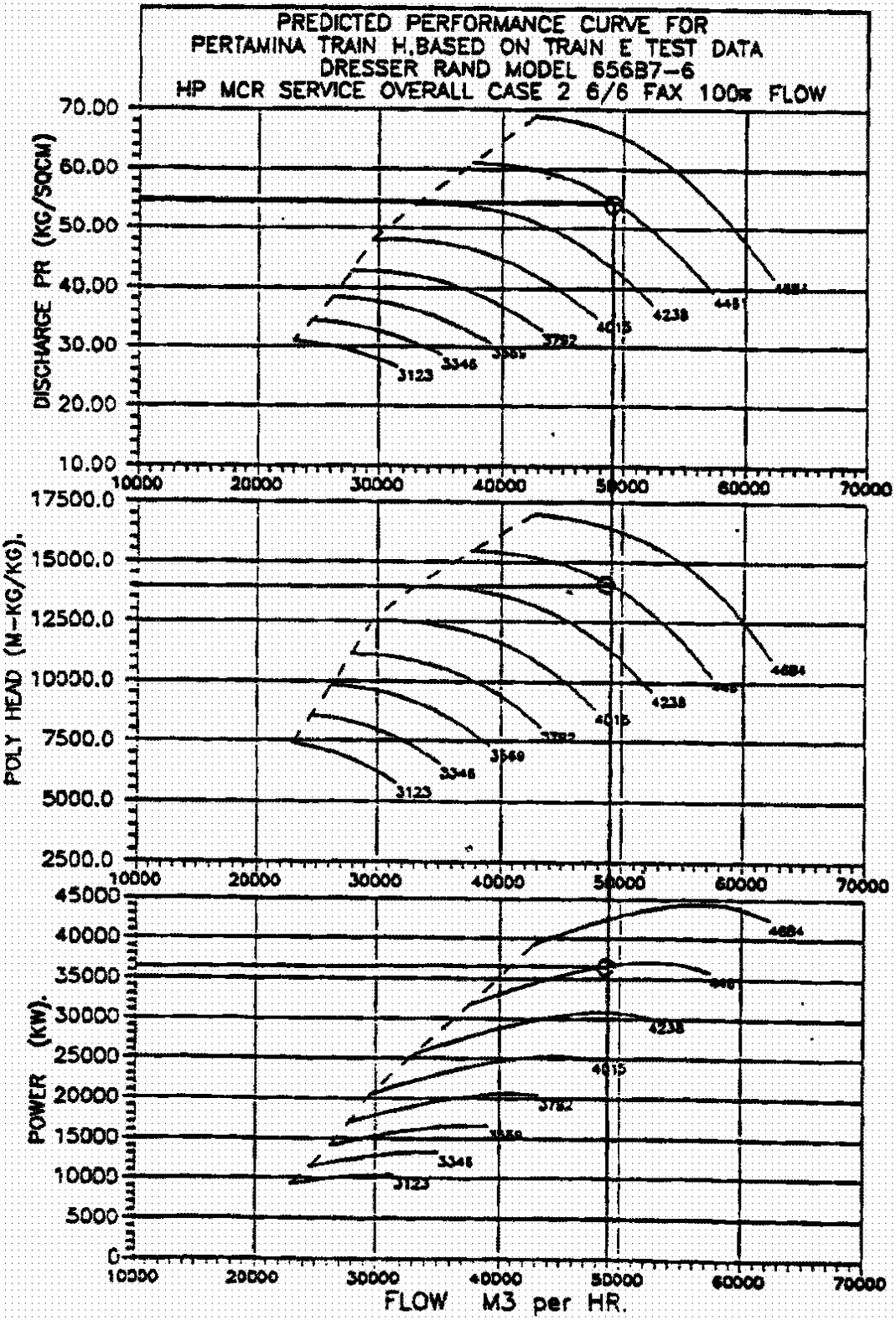
JOB NO. *95038/6413* ITEM NO. *H4K-3*
P.O. NO. _____ DATE *16 April 1997*
REQUISITION NO. *PGH-MR-K-003*
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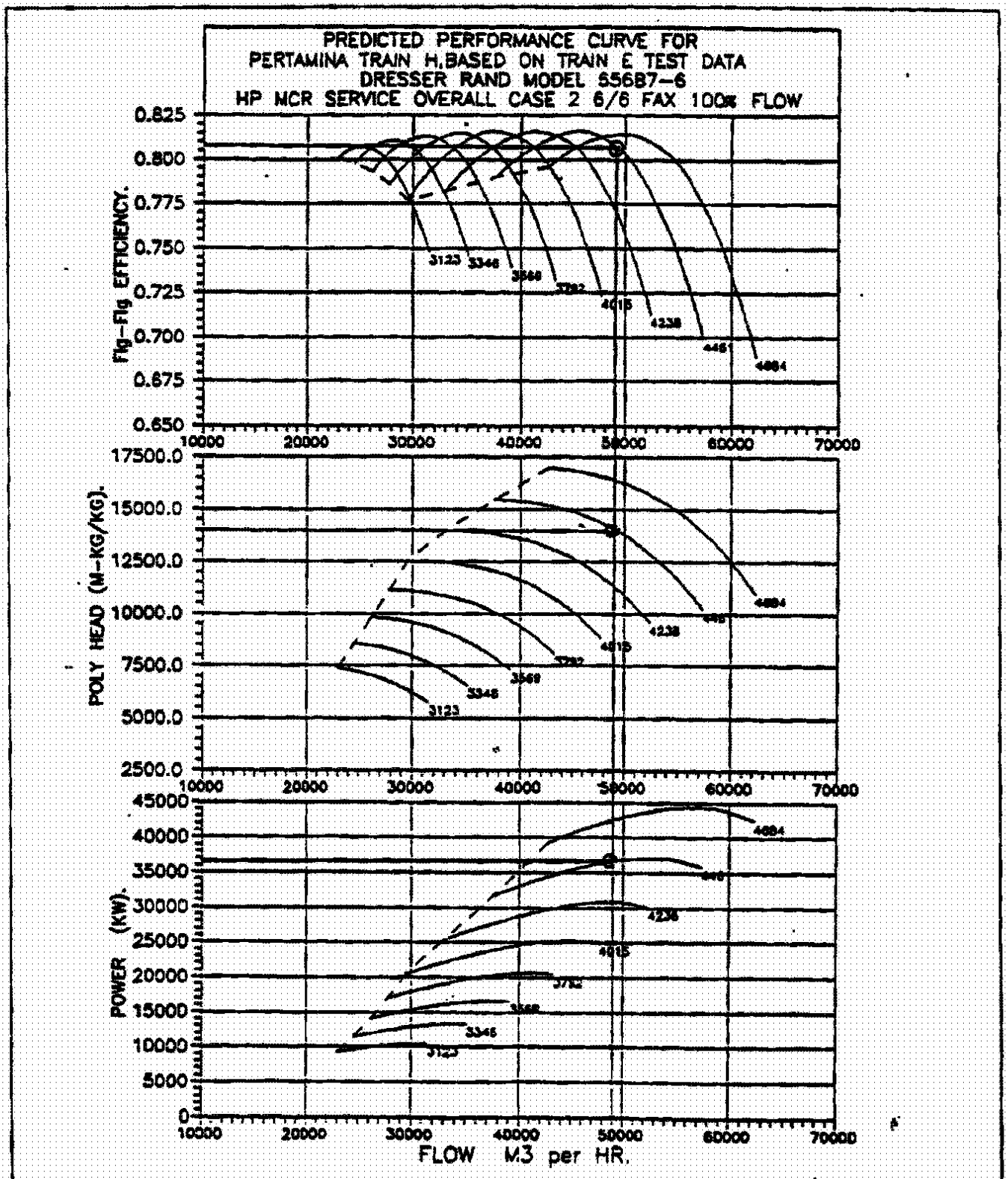
DATE: 6- 5-96	TTB	CURVE No.:
N5-IN0224		HPMCR20

DRESSER-RAND

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CENTRIFUGAL COMPRESSOR
DATA SHEET
SI (METRIC) UNITS

JOB NO. *95038/6413* ITEM NO. *H4K-3*
P.O. NO. _____ DATE *16 April 1997*
REQUISITION NO. *PGH-MR-K-003*
DOC. NO. *DS-H04-K-003*



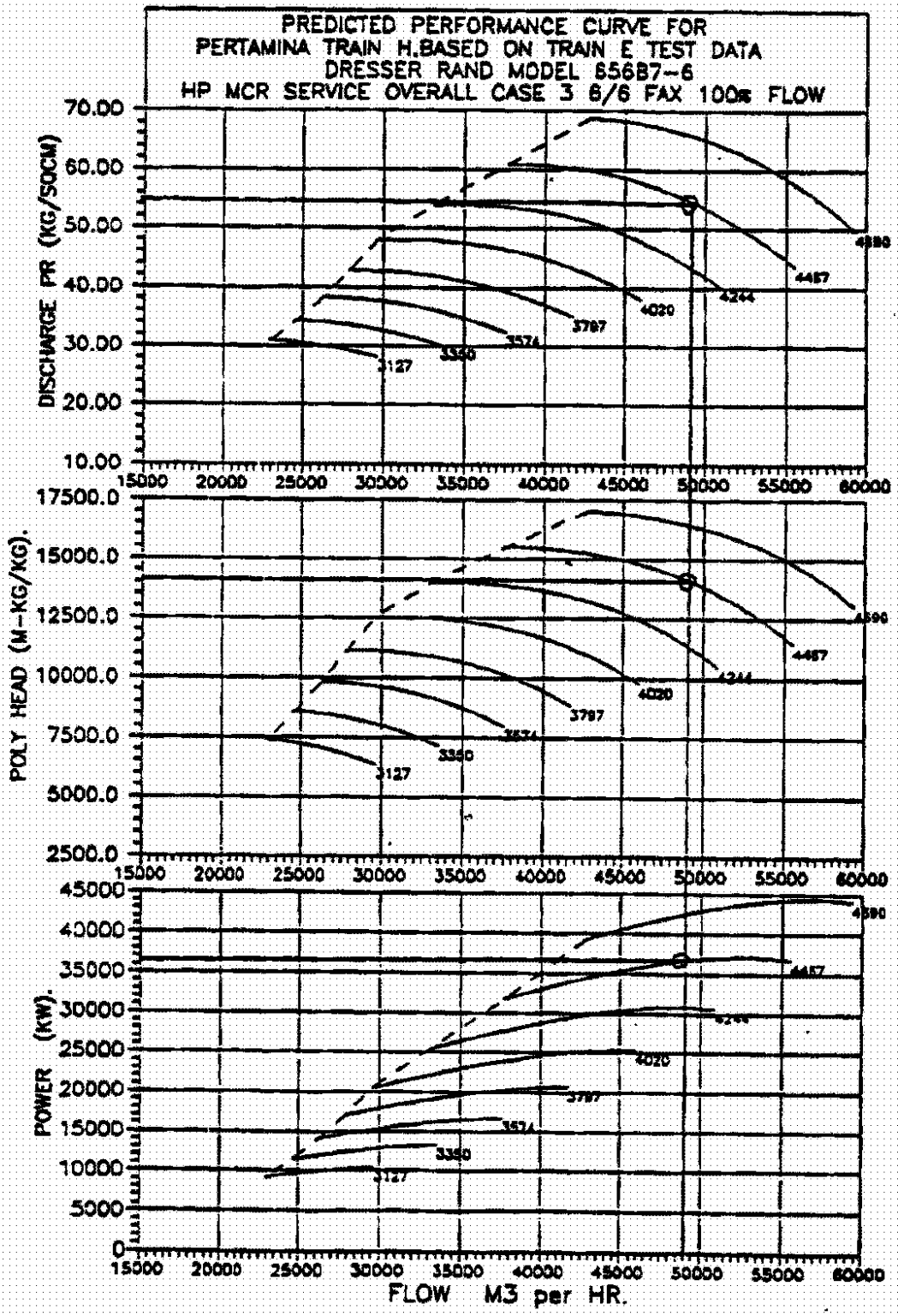
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CENTRIFUGAL COMPRESSOR
DATA SHEET
SI (METRIC) UNITS

JOB NO. *95038/6413* ITEM NO. *H4K-3*
P.O. NO. _____ DATE *16 April 1997*
REQUISITION NO. *PGH-MR-K-003*
DOC. NO. *DS-H04-K-003*



DATE: 5- 6-96	TTB	CURVE No.:
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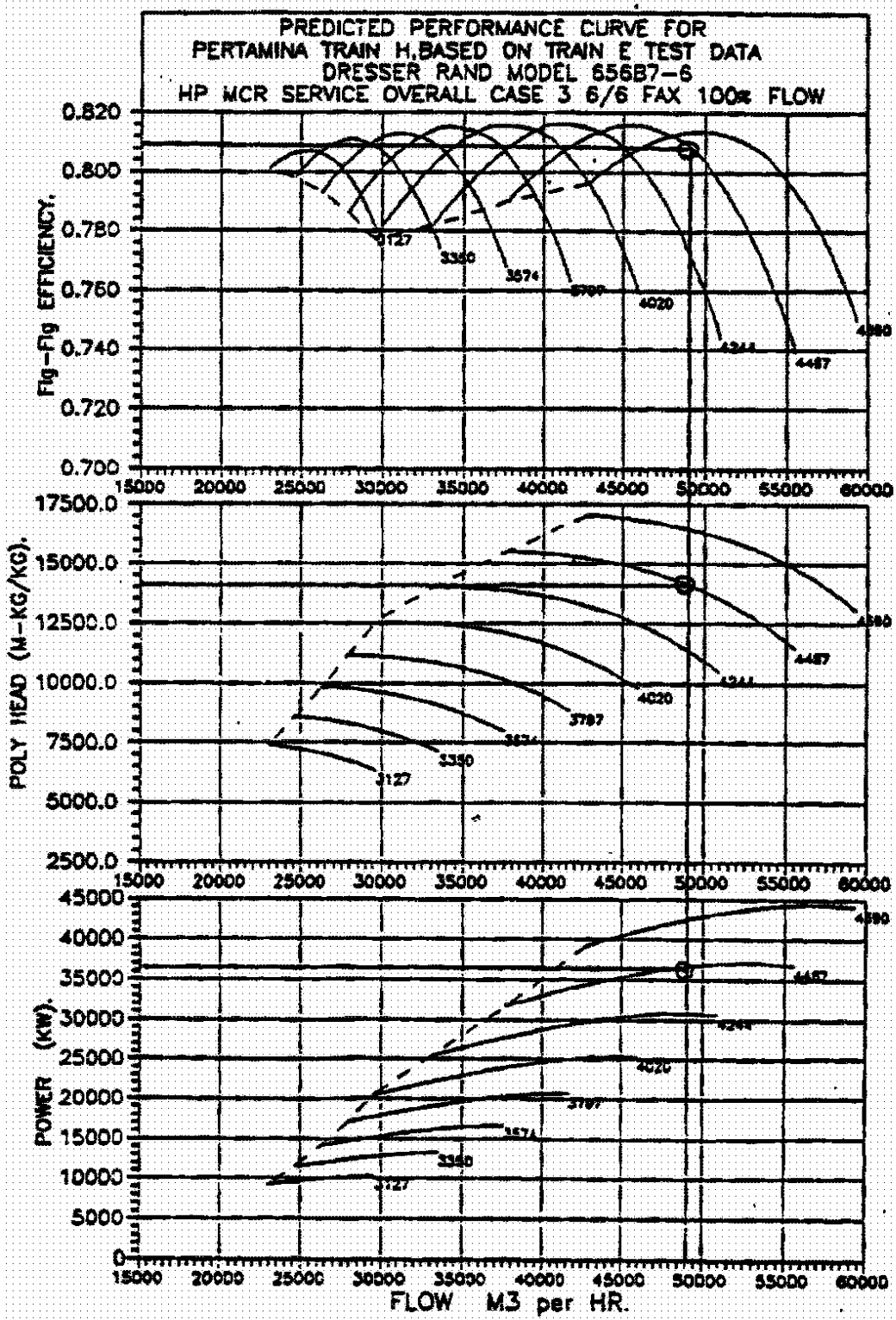
DRESSER-RAND

IKPT

KELLOGG

CENTRIFUGAL COMPRESSOR
DATA SHEET
SI (METRIC) UNITS

JOB NO. *95038/6413* ITEM NO. *H4K-3*
P.O. NO. _____ DATE *16 April 1997*
REQUISITION NO. *PGH-MR-K-003*
DOC. NO. *DS-H04-K-003*



DATE: 5- 6-96	TTB	CURVE No.:
N5-IN0224		HPMCR30A

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IKPT KELLOGG

CENTRIFUGAL COMPRESSOR DATA SHEET SI (METRIC) UNITS

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P.O. NO. DATE 16 April 1997
REQUISITION NO. PGH-MR-K-003
DOC. NO. DS-H04-K-003

OPERATING CONDITIONS (CONTD) (3.1.2 & 3.1.3)

GAS ANALYSIS ◉ MOL % ○	MW	DESIGN & GUARANTEE CASE 1	CASE 2	CASE 3	REMARKS (2.11.1.3 & 2.11.1.8)
Air	28.966				
Oxygen	32.000				
Nitrogen	28.016	3.60	3.45	3.32	
Water Vapor	18.016				
Carbon Monoxide	28.010				
Carbon Dioxide	44.010				
Hydrogen Sulfide	34.076				
Hydrogen	2.016				
Methane	16.042	45.16	44.75	46.01	
Ethylene	28.052				
Ethane	30.068	41.16	42.37	39.98	
Propylene	42.078				
Propane	44.094	10.08	9.43	10.69	
iso Butane	58.120				
n Butane	58.120				
iso Pentane	72.146				
n Pentane	72.146				
Hexane Plus					
TOTAL		100	100	100	
Avg Mol Wt		25.075	25.044	25.046	

LOCATION
○ Indoor ○ Heated ◉ Under Roof
◉ Outdoor ○ Unheated ◉ Partial Sides
○ Grade ◉ Mezzanine ○
◉ Electrical Area Class. (2.1.15) Cl I Gr D Div 2
○ Winterization Reqd (2.1.9) ◉ Tropicalization Reqd (3.4.6.6)

NOISE SPECIFICATIONS (2.1.10)
◉ Applicable to Machine
See Specification SPEC. A-9
○ Applicable to Neighborhood
See Specification
Acoustic Housing : ○ YES ◉ NO

SITE DATA
◉ Elevation 14 m ◉ Barometer 1.03 Kg/cm2
◉ Range of Ambient Temp.° C :

	DRY BULB	WET BULB
Site Rated	30°C	28°C
Normal	26°C	24°C
Max.	37°C	-
Min.	21°C	-

APPLICABLE SPECIFICATIONS API Std 617
◉ Vendor Having Unit Responsibility (2.9.1.7)
◉ Governing Specification
SPEC. K-1 (C.2.7.1)

PAINTING
○ Manufacturer's ◉ Other SPEC. X-2 & PGG-IS-X-1002 REV. 2

UNUSUAL CONDITIONS ○ Dust ○ Fumes
◉ Other (2.1.9)
SALT LADEN ATMOSPHERE

SHIPMENT (4.4.1)
○ Domestic ◉ Export ◉ Export Boxing Reqd
◉ Outdoor Storage More Than 6 Months (4.4.1)
Spare Rotor Assembly Packaged for (4.4.3.10)
○ Horizontal Storage ○ Vertical Storage

REMARKS

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CENTRIFUGAL COMPRESSOR DATA SHEET SI (METRIC) UNITS

JOB NO. 95038/6413 ITEM NO. H4K-3
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CONSTRUCTION FEATURES

☒ SPEEDS
Max Cont 4835 rpm Trip 5319 rpm
Max. Tip Speeds : 214 m/s @ rated Speed
232 m/s Max. Cont Speed

☒ LATERAL CRITICAL SPEEDS (DAMPED)
First Critical 2800 rpm Mode
Second Critical 7900 rpm Mode
Third Critical rpm Mode
Fourth Critical rpm Mode
1 • Lateral Analysis Required
• Undamped Stiffness Map Required
• Train Torsional Analysis Required
(Turbine Driven Train) (2.9.4.5)

☐ TORSIONAL CRITICAL SPEEDS
First Critical rpm
Second Critical rpm
Thrid Critical rpm
Fourth Critical rpm

☒ VIBRATION
Allowable Test Level (pk-to-pk) 38.1 Microns

ROTATION, VIEWED FROM DRIVEN END

• MATERIALS INSPECTION REQUIREMENTS (4.2.2)
• Special Charpy Testing (2.11.3) Yes
• Radiography Required for (See Note 2 Pg 14)
3 • Magnetic Partical Required for (See Note 2 Pg 14)
Liquid Penetrant Required for

1 ☒ CASING
Model 656 B7 - 6
Casing Split RADIAL
Material ASTM A 516 Gr 70
Thickness, mm 95.25 Corr. Allow., mm 3.18
Max Working Press 62.2 kg/cm2G
Max Design Press. 70 kg/cm2G
Test Press , Kg/cm2G: Helium N/A Hydro 105
Max. Oper Temp (1) 204.4 °C Min. Oper Temp. - 17.8 °C
Max No. of Impellers for casing SIX
Max. Casing Capacity, m3/h 70000
3 Radiograph Quality • Yes No
Casing Split Sealing O - RINGS
• System Relief Valve Set Pt (2.2.4) 62.2 kg/cm2G

☒ DIAPHRAGM
Material CAST IRON

☒ IMPELLERS
No. 6 Diameters 4 @ 914
4 No. Vanes Each Impeller 19, 19, 19, 17, 17, 17
(1) VENDOR SHALL CONFIRM BASED ON CONTINUOUS RECYCLE CASE

Type (Open, Enclosed, etc) ENCLOSED
Type Fabrication WELDED
Material AISI 4330
Min. Yield Strength, Kg/cm2 6329
Brinnel Hardness : Max. 302 Min. 241
Smallest Tip Internal Width, mm 46
Max. Mach No. @ Impeller Eye .60
Max. Impeller Head @ Rated Speed, m 2703

☒ SHAFT
Material AISI 4340
Diam @ Impellers, mm 225 Diam @ Coupling, mm 177.8
Shaft End : ☒ Tapered Degree of Angle
Max. Yield Strength, bar 8620
Shaft Hardness (BNH) (Rc) 293 - 331
Stress at Coupling, bar 722

☒ BALANCE PISTON
Material 4140 Area 177.742 mm2
Fixation Method INTERFERENCE FIT

☒ SHAFT SLEEVES
At Intersig Close Clearance Points Matl 410 SS
At Shaft Seals Matl MONEL "B" WITH COLMONOY OVERLAY

☒ LABYRINTHS
Interstage :
Type KNIFE EDGE Material ALUMINIUM
Balance Piston :
Type KNIFE EDGE Material ALUMINIUM

SHAFT SEALS
◉ Seal Type (2.8.3) OILFILM
◉ Settling Out Pressure (400PSIG) 281 kg/cm2G
○ Special Operation (2.8.1)
○ Supplemental Device Required for Contact Seals (2.8.3.2)
Type
○ Buffer Gas System Required (2.8.7)
○ Type Buffer Gas
○ Buffer Gas Control System Schematic by Vendor
☐ Type Seals
☒ Inner Oil Leakage Guar. L/day/seal 352 TOTAL
Buffer Gas Required :
☐ Air-Run-in ☐ Other
☐ Buffer Gas Flow (per Seal) :
Norm : kg/min @ bar Δ P
Max. : kg/min @ bar Δ P.

☒ BEARING HOUSING CONSTRUCTION
Type (Separate, Integral) SEPARATE Split HORIZONTAL
Material C.S ASTM A 36
Seperation Seal, Type Split
Material

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CONSTRUCTION FEATURES (CONTD)

BEARINGS AND BEARING HOUSINGS

RADIAL	INLET	EXHAUST
☒ Type	*Tilting Pad*	*Tilting Pad*
☒ Manufacturer	*D-R*	*D-R*
☒ Length, mm	*71.1*	*71.1*
☒ Shaft Diam, mm	*177.8*	*177.8*
☒ Unit Load (Act. / Allow.), bar	*6.1 / 34.5*	*6.1 / 34.5*
☒ Base Material	*ASTM A36*	*ASTM A36*
☒ Babbit Thickness, mm	*1.524*	*1.524*
☒ No. Pads	*FIVE*	*FIVE*
☒ Load Between / On Pad	*ON*	*ON*
☒ Pivot : Center / Offset, %	*CENTER*	*CENTER*
☒ *Bearing Span*	*2422.5*	
☐		

THRUST	ACTIVE	INACTIVE
☒ Type	*Tilting Pad*	*Tilting Pad*
☒ Manufacturer	*D-R*	*D-R*
☒ Unit Loading, Max. bar	*9.4*	*9.4*
☒ Unit Load, Ult bar	*69*	*69*
☒ Area, mm2	*48387*	*48387*
☒ No. Pads	*8*	*8*
☒ Pivot : Center / Offset, %	*55*	*55*
☒ Pad Base Material	*ASTM A36*	*ASTM A36*

Lubrication : ● Flooded ○ Directed
Thrust Collar : ○ Integral ● Replaceable
Material *FORGED STEEL*

BEARING TEMPERATURE DEVICES

See Attached API 670 Data Sheet
Thermistors
Type - Pos Temp Coeff - Neg Temp Coeff
Temp Switch & Indicator by : - Purch - Mfr
● Thermocouples
● Selector Switch & Indicator by : ✓ Purch - Mfr
Resistance Temp Detectors
Resistance Matl - Ohms
Selector Switch & Indicator by : - Purch - Mfr
● Location-Journal Bearing
No. - Each Pad - Every Other Pad *4** per Brg
Other *(*) Include 2 Spares*
● Location-Thrust Bearing
No. - Each Pad - Every Other Pad *6*** per Brg
Other *(**) Include 2 Spares*
No (Inact) - Each Pad - Every Other Pad *2* per Brg
Other
● Monitor Supplied by (3.4.7.4) PURCHASER
Location - Enclosure
Mfr - Model
Scale Range - ● Alarm ☒ Set @ *110* °C
3 Shutdown Set @ *121* °C Time Delay - sec

VIBRATION DETECTORS

See Attached API 670 Data Sheet
● Type *PROXIMITY* ✓ Model *3300*
● Mfr *BENTLY NEVADA*
● No. at Each Shaft Bearing *4**** Total No. *8*
● Oscillator-Detectors Supplied by *COMPRESSOR MFR.*
● Mfr *BENTLY NEVADA* ✓ Model *3300*
Monitor Supplied by (3.4.7.2) *PUCHASER*
● Location *CCR* Enclosure
● Mfr *BENTLY NEVADA* ✓ Model *3300*
Scale Range ● Alarm ✓ Set @ *102* mm
● Shutdown ☒ Set @ *132* mm ○ Time Delay sec 3

AXIAL POSITION DETECTOR

See Attached API 670 Data Sheet
● Type *PROXIMITY* ☒ Model *3300*
● Mfr *BENTLY NEVADA* ● No. Required *3*****
● Oscillator-Demodulator Supplied by *COMPR. MFR.*
● Mfr *BENTLY NEVADA* ✓ Model *3300*
Monitor Supplied by (3.4.7.2) *PURCHASER*
● Location *CCR* Enclosure
● Mfr *BENTLY NEVADA* ✓ Model *3300*
Scale Range ● Alarm ☒ Set @ *127* mm
● Shutdown ☒ Set @ *254* mm ○ Time Delay sec 3

CASING CONNECTIONS

CONNECTION	● DESIGN APPROVAL REQD (2.11.2.10) 0	☒ SIZE	☒ FACING	● POSITION (2.4.2.1)	◉ FLANGED OR STUDDED (2.4.2.1)	◉ MATING FLG & GASKED BY VENDOR (2.4.2.3)	☒ GAS VELOCITY @ MCS m/s Rtd / Max. Allow.
Inlet	*Yes*	*24*	*ANSI 400# RF*	*DOWN*	*FLANGED*	*Yes*	*47.7*
Discharge	*Yes*	*20*	*ANSI 400# RF*	*DOWN*	*FLANGED*	*Yes*	*25.8*

(***) INCLUDES 2 SPARES INSTALLED
(****) INCLUDES 1 SPARE INSTALLED

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☐ OTHER CONNECTONS

SERVICE	NO	SIZE	TYPE
Lube Oil Inlet			
Lube Oil Outlet			
Seal Oil Inlet			
Seal Oil Outlet			
Casing Drains			
Stage Drains			
Vents			
Cooling Water			
Pressure			
Temperature			
Purge for :			
Brg Housing			
Between Brg & Seal			
Buffer Gas			
Solvent Injection			

◉ Individual Stage Drains Required (2.4.3.2)
○ Valved & Blinded
◉ Valved & Blinded & Manifold

☒ ALLOWABLE PIPING FORCES AND MOMENTS

	INLET		DISCHARGE			
⚠1	FORCE N	MOMT N.m	FORCE N	MOMT N.m	FORCE N	MOMT N.m
AXIAL	PER	API	617	6TH	ED.	
VERTICAL						
HORIZ 90°						
	FORCE N	MOMT N.m	FORCE N	MOMT N.m	FORCE N	MOMT N.m
AXIAL	PER	API	617	6TH	ED.	
VERTICAL						
HORIZ 90°						

○ ACCELEROMETER
○ See Attached API 670 Data Sheet
○ Type ☐ Model
○ Mfr No. Required
○ Location
○ Oscillator-Demodulators Supplied by
○ Mfr ☐ Model
○ Monitor Supplied by (3.4.7.6)
○ Location Enclosure
○ Mfr ☐ Model
☐ Scale Range ○ Alarm : ☐ Set @ mm/s2
○ Shutdown ☐ Set @ mm/s2 ○ Time Dela sec

ACCESSORIES

COUPLING AND GUARD 3.2

NOTE : See Rotating Elements - Shaft Ends
○ See API 671 Data Sheet
Coupling Furnished by *Compressor Mfr*
Manufacturer *Lucas (Bendix)* Type *Countoured Diaphragm Flexible* Model *74 F522A*
Coupling Guard Furnished by *Compressor Mfr* No. of Vent Conn : *1* Size, mm *177.8*
Type . ◉ Fully Enclosed ○ Semi-open ○ Oth
Coupling Details :
☐ Max. OD mm
☐ Hub Weight kg
☐ Spacer Length mm
☐ Spacer Weight kg

◉ Vendor Mount Half Coupling
○ Idling Adapter / Solo Plate Reqd (3.2.4)
Lubricating Requirements :
◉ Nonlube ○ Grease ○ Cont Oil Lube ○ Other
Quantity per Hub kg or m3/h
◉ Hydraulic coupling removal / attachment KIT
(Adapter + Hydraulic Pump)

MOUNTING PLATES

◉ Baseplates : Furnished by (3.3.1.1) *Compressor Mfr*
◉ Compressor Only (3.3.2.1) ○ Driver ○ Gear
○ Other
○ Drip Trim ○ Leveling Pads (3.3.2.2)
○ Column Mounting (3.3.2.3)
○ Subsoleplates Reqd (3.3.3.2)
☒ Stainless Steel Shim Thickness mm
◉ Primer for Epoxy Grout Reqd (3.3.1.2.9)
Type *Catalyzed Epoxy Primer*

○ Soleplates : Furnished by
☐ Thickness mm
○ Subsoleplates Reqd (3.3.3.2)
○ Leveling (Chock) Blocks Reqd
☐ Stainless Steel Shim Thicknes
○ Driver ○ Gear ○ Compressor
○ Primer for Epoxy Grout Reqd (3.3.1.2.9)
Type

REMARKS

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UTILITIES

◉ UTILITY CONDITIONS

	DRIVERS		HEATING	
STEAM				
Inlet Min. 17.6	kg/cm2	°C	barg	°C
Norm. 17.6	kg/cm2	°C	barg	°C
Max. 19.3	kg/cm2	°C	barg	°C
Exhaust Min. 3.9	kg/cm2	°C	barg	°C
Norm. 3.9	kg/cm2	°C	barg	°C
Max. 3.9	kg/cm2	°C	barg	°C

ELECTRICITY

	DRIVERS	HEATING	CONTROL	SHUTDOWN
Voltage	460	240	120	120
Hertz	50	50	50	50
Phase	3	1	1	1

COOLING WATER

Temp : Inlet	34	°C	Max. Return	46	°C
Press. : Norm.	2.1	Kg/cm2G	Design	8.8	Kg/cm2G
Min. Return	1.1	Kg/cm2G	Max. Allow. ΔP	1.0	Kg/cm2G
Water Source			SEA WATER		

INSTRUMENT AIR

Max. Press.	9.0	Kg/cm2G	Normal Press.	7.0	Kg/cm2G
			Min. Press.	4.6	Kg/cm2G

SHOP INSPECTION AND TEST (4.1.4)

	REQD	WITNESS	OBSER'D
Shaft Runout	◉	◉	
Shop Inspection	◉	◉	
Cleanlines (4.2.1.5)	○		
QC Program Review	◉		
Hydrostatic	◉	◉	○
Impeller Overspeed	◉	◉	○
Mechanical Run *with shop driver*	◉	◉	○
☒ Contract Coupling ☐ Idling Adaptor(s)			
☒ Contract Probes ☐ Shop Probes			
Vary Lube & Seal Oil Pressures and Tempeatures (4.3.4.2.5)	◉		
Polar Form Vibration Data (4.3.4.3.3)	◉		
Tape Record Vibration Data (4.3.4.3.6)	◉		
Tape Data to Purchaser (4.3.4.3.7)	◉		
Shaft End Seal Inspection (4.3.4.4.1)	◉	◉	
Gas Leak Test Disch Press (4.3.5.3)	◉	◉	○
○ Before ◉ After Mech Run			
Performance Test (Gas) (4.3.6.1)	◉	◉	○
Complete Unit Test (4.3.6.2)	○	○	○
Torsional Vibration Meas (4.3.6.2)	○	○	○
Tandem Test (4.3.6.3)	○	○	○
Gear Test (4.3.6.4)	○	○	○
Helium Leak Test (4.3.6.5)	○	○	○
Sound Level Test (4.3.6.6)	◉	◉	○
Full Load/Speed/Press. Test (4.3.6.9)	○	○	○
Hydraulic Coupling Inspection (4.3.6.10)	◉	◉	○
Certified Copies of Test Data	◉	○	○
Disassy. / Reassy Compressor After Test (3)	◉	◉	○
Preparation for Shipment	◉	◉	○
Residual Elect / Mech Run Out	◉	◉	○
Bearing / Seal Disassembly	◉	◉	
Unbalance Response Verification	◉	◉	

☐ TOTAL UTILITY CONSUMPTION

Cooling Water	m3/h
Steam, Normal	Kg/h
Steam, Max.	Kg/h
Instrument Air	mm3/h
Horsepower (Driver)	kW
Horsepower (Auxiliaries)	kW
Heaters	kW
Purge (Air or N2)	mm3/h

MISCELLANEOUS

☒ Recommended Straight Run of Pipe Diameters
◉ Before Suction 3 - 4
◉ Vendor's Review & Comments on Purchaser's Piping & Foundation (3.5.3.2)
○ Compressor to be Suitable for Field Run-in on Air
○ Provision for Liquid Injection (2.1.11)

◉ Vendor's Review & Comments on Purchaser' Control System (3.4.1.1)
○ Extent of Process Piping by Vendor (3.5.3.1)
○ Shop Fitup of Vendor Process Piping (4.4.3.11)
○ Welding Hardness Testing (4.2.1.6)
○ Auxiliary Equipment Motors Explosionproof (3.1.8)
◉ Torsional & Lateral Critical Analysis Report Req'd.

☒ WEIGHTS, kg

Compr 26900 Gear ____ Driver 47200 Base 4900
Rotors : Compr 1580 Driver ____ Base ____
Compr Upper Case ____
Sour Seal Oil Traps ____
L. O. Console 41500 ____ S. O. Console
Overhead Seal Oil Tanks ____
Max. for Maintenance (Identify) 11600
Total Shipping Weight 31,700 Comp. 47,200 S.Turbine

☒ SPACE REQUIREMENTS, m

Complete Unit : L 12400 W 4100 H 4900
L. O. Console : L 11500 W 5600 H 2600
S. O. Console : L ____ W ____ H ____
Sour Seal Oil Tanks ____
Overhead Seal Oil Tanks ____

SAFETY CONTROLS

Control Logic : c
- Relay - Single - Dual - 2 of 3
- PLC - Dual - 2 Out of 3

Vendor to Privide :
- Total System - Local Components

PLC :
Interface - Mfr

REMARKS

1. UTILITY CONDITIONS, REFER TO SH-ED-PE-03 (1)
2. OTHER SHOP INSPECTION AND TESTS SHALL BE BE CONDUCTED AS PER PGH-QC-A-002
3. IF UNSATISFACTORY RUN TEST (2)

IKPT KELLOGG

CENTRIFUGAL COMPRESSOR
DATA SHEET
SI (METRIC) UNITS

JOB NO. 95038/6413 ITEM NO. H4K-3
P.O. NO. _____ DATE 16 April 1997
REQUISITION NO. PGH-MR-K-003
DOC. NO. DS-H04-K-003

SPECIAL NOTES :

(1) THE ACCEPTANCE CRITERIA FOR THE SHAFT HORSE POWER INCLUDING ALL LOSSES AT THE GUARANTEE POINT (CASE 1) SHALL NOT EXCEED 104% OF THE SPECIFIED VALUE DURING THE PTC-10 SHOP PERFORMANCE TEST.

(2) MATERIAL TEST, SHOP INSPECTION AND TEST REQUIREMENTS OTHER THAN THOSE SPECIFIED IN THIS DATA SHEET SHALL BE REFERRED TO PGH-QC-A-002 " SHOP INSPECTION AND TEST PLAN".

(3) COMPRESSOR AND TURBINE VENDOR SHALL DEMONSTRATE INTERCHANGEABILITY OF THE COMPLETE BUNDLE (ROTOR, DIAPHRAGMS, ETC) WITH EXISTING 4K-3 CASES (IE. IN THE EXISTING TRAIN G) . VENDOR TO PROVIDE A DIMENSIONAL RECORD TO HCJV (PURCHASER) FOR REVIEW AND APPROVAL.

(4) SEAL AND BEARING PARTS MUST BE INTERCHANGEABLE WITH THE EXISTING 4K-3's TRAIN-G

(5) SHAFT COUPLING SHALL BE RATED ACCORDING TO THE POWER REQUIREMENTS SPECIFIED ON PAGE 2.

(6) OTHERS:

a. TORSIONAL AND LATERAL CRITICAL ANALYSIS REPORT REQUIRED WITH COMPLETE ROTOR DYNAMICS AND BEARING STABILITY REPORT PER SPECIFICATION

b. VENDOR TO TERMINATE ALL WIRING AT TERMINAL BOXES AT EDGE OF SKID

c. OPTICAL ALIGMENT PADS (BENCH MARKS) REQUIRED ON COMPRESSOR / DRIVER SKID

d. DELETED

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KELLOGG

COUPLING (API 671-2ND)
DATA SHEET
SI UNITS

JOB NO. 95038 / 6413 ITEM NO H4K-3
PURCHASE ORDER NO. PGH-PO-K-003
REQUISITION NO PGH-MR-K-003
REVISION NO. 4 DATE 16/4/97
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APPLICABLE TO: ○ PROPOSAL ● PURCHASE ○ AS BUILT DATE 16/4/97 REVISION 4

● FOR BONTANG LNG EXPANSION PROJECT, TRAIN - H NO.REQUIRED 1 (ONE)
SITE BONTANG, EAST KALIMANTAN, INDONESIA SERVICE HP MR COMPRESSOR

● DRIVER TYPE STEAM TURBINE MANUFACTURER GENERAL ELECTRIC MODEL DRV 1034
SERIAL NO TAG. NO.
DRIVEN UNIT COMPRESSOR MANUFACTURER DRESSER RAND MODEL 656 B7-6
SERIAL NO TAG NO.

■ COUPLING TYPE CONTOURED DIAPRHRAGM FLEXIBLE
MANUFACTURER LUCAS (BENDIX) MODEL 74F522A SIZE
ASSEMBLY DRAWING NO

NOTE: INFORMATION TO BE COMPLETED, ○ BY PURCHASER; ☐ BY COUPLING MANUFACTURER.

OPERATING CONDITIONS

● POWER TRANSMITTED (kW). NORMAL 36081 MAXIMUM CONTINUOUS
● SPEED (RPM) MINIMUM ALLOWABLE 3556 NORMAL 4445 MAXIMUM CONTINUOUS 4835 TRIP 5319
○ TORQUE (N-m) MAXIMUM CONTINUOUS MAXIMUM TRANSIENT
○ FREQUENCY OF TRANSIENTS (EVENTS/TIME) (2.1.1)
○ ADDITIONAL ANF EXCITING FREQUENCY (2.3.4.C)
● MINIMUM REQUIRED EXPERIENCE FACTOR, SF (2.1.1) 1.75 ☐ ACTUAL SF
● AMBIENT TEMPERATURE (°C) MAXIMUM 37 MINIMUM 21
ENVIRONMENT (2.6.5) (2.1.6) ○ DUST ○ HYDROGEN SULFIDE ● SALT LADEN ATMOSPHERE

LUBRICATION

○ CONTINUOUS (2.2.7 & 8) ○ FILTRATION MICRONS ○ VISCOSITY (cP) @ °C
○ BATCH (2.2.7) ● NONLUBRICATED ○ PRESSURE (kPa) (BAR) NORMAL MAXIMUM
○ GREASE-PACKED ○ TEMP. (°C) NORMAL MAXIMUM
○ OILFILLED ☐ FLOW (m³/h) NORMAL MAXIMUM

COUPLING DATA

● SHAFT SEPARATION AT OPERATING TEMP. (mm B.S.E.) (1) THERMAL GROWTH (mm) FROM °C AMBIENT
○ MOTOR ROTOR FLOAT (mm) ○ LIMITED END FLOAT (mm) (2.2.3) ○ ELECTRICALLY INSULATED (2.1.12)
○ MARINE TYPE REQUIRED (2.2.2) ○ FLEX-HUB TYPE REQUIRED (2.2.2)
○ REQUIRED MISALIGNMENT CAPABILITY (2.1.2)
STEADY STATE ANGULAR (DEG.) PARALLEL OFFSET (mm) AXIAL(mm)
TRANSIENT ANGULAR (DEG.) PARALLEL OFFSET (mm) AXIAL(mm)
○ DYNAMIC BALANCE (2.5.2.1) ● COMPONENT BALANCE & ASSEMBLY CHECK BALANCE (2.5.2.1) (2.5.2.3)
○ ASSEMBLY BALANCE (2.5.2.4) ○ REPEATABILITY TEST AFTER ASSEMBLY BALANCE (2.5.2.5)
○ RESIDUAL UNBALANCE CHECK OF COMPONENTS (2.5.2.2)
☐ DESIGN RATING (kW/100 RPM) (2.1.1) ☐ MAXIMUM CONTINUOUS TORQUE (N-m) (5.2.5.6c)
☐ BOLTING TORQUE (N-m) (2.1.11.5) ☐ LUBRICATED ☐ DRY
☐ MAXIMUM ALLOWABLE MISALIGNMENT (5.2.5.6a)
STEADY STATE ANGULAR (DEG.) PARALLEL OFFSET (mm) AXIAL(mm)
TRANSIENT ANGULAR (DEG.) PARALLEL OFFSET (mm) AXIAL(mm)
☐ MAXIMUM ALLOWABLE RESIDUAL UNBALANCE (g-mm) DRIVER END DRIVEN UNIT END
☐ MAXIMUM ACTUAL RESIDUAL UNBALANCE (g-mm) DRIVER END DRIVEN UNIT END
☐ TORSIONAL STIFFNESS (kg-mm/RAD) (5.2.5.2b) WR²(kg-m²) (5.2.5.2b)
☐ FLEXIBLE-ELEMENT COUPLING INITIAL DEFLECTION (mm) ☐ PRESTRETCH ☐ COMPRESSION
☐ FLEXIBLE-ELEMENT COUPLING CALCULATED AXIAL NATURAL FREQUENCY
○ TEST OF ANF (4.4.1) ☐ ACTUAL ANF

(1) DUPLICATE TO TRAIN - G

NOTE VENDOR TO COMPLETE THE DATA IN THE BLANK CELLS

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COUPLING (API 671-2ND) DATA SHEET SI UNITS

JOB NO. 95038 / 6413 ITEM NO HAR-3
REVISION NO. 4 DATE 16/4/97
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COUPLING DATA (CONTINUED)

● REQUIRED CURVES (5.2.4.1) (5.2.4.2)
- ● AXIAL FORCE VS AXIAL DEFLECTION
- ● AXIAL FORCE VS TORQUE
- ● BENDING MOMENT VS AXIAL DEFLECTION
- ● BENDING MOMENT VS ANGULAR MISALIGNMENT
- ○ WINDAGE LOSS VS COUPLING SPEED

MATERIALS (2.6)

HUB/FLANGE	DRIVE END MATERIALS	DRIVEN END MATERIALS
SPACER		
SLEEVE		
FLEXIBLE-ELEMENT		
FLEXIBLE-ELEMENT GUARD		
BOLTS		
NUTS		

● FLEXIBLE-ELEMENT PROTECTIVE COATING (2.6.4) ● VENDOR STANDARD ○
☐ MATERIAL AND WELD HARDNESS TESTS REQUIRED (4.3.1.3)
☐ INTERNAL TEETH HARDNESS (ROCKWELL C) (2.2.6) DRIVE END ACTUAL DRIVEN END ACTUAL
☐ EXTERNAL TEETH HARDNESS (ROCKWELL C) (2.2.6) DRIVE END ACTUAL DRIVEN END ACTUAL

COUPLING HUB MACHINING

	DRIVE END	DRIVEN END
● TYPE (INTEGRAL, CYLINDRICAL, TAPER) (2.1.4) (2.1.8.1)	INTEGRAL	
● KEYED OR HYDRAULICALLY FITTED (2.1.8.1)		HYDRAULIC FIT
○ TAPER (1 DEG. I.A., 1/2 IN. PER FT., 3/4 IN. PER FT.) (2.1.8.1.1) (2.1.8.1.2) (2.1.8.1.3)		
☐ KEYWAY DIMENSIONS AND NUMBER (2.1.8.2)		
☐ NOMINAL BORE DIAMETER		
☐ INTERFERENCE FIT (mm) MAX./MIN. (2.1.8.1)	/	/
○ PULLER HOLES		
○ BALANCING HOLES (2.5.3)		

COUPLING GUARD (APPENDIX E)

- ○ COORDINATOR (E.2.1)
- ● FLANGED CYLINDRICAL (E.4.1) ○ BASE MOUNT (E.3.2)
- ○ AIR TIGHT (E.4.7.1) ● OIL TIGHT (E.2.7)
- ● SPARK RESISTANT (E.2.9)
- ○ TRANSPARENT WINDOW FOR EACH OIL SPRAY POINT (E.4.6.4)
- ● VENT CONNECTION (E.4.3) (1)
  - ○ 1 INCH NPT WITH FILTER-BREATHER
  - ○ 1 INCH FLANGED, RATING & FACING
- ○ PURGE GAS ○ DRY AIR PURGE (E.4.4)
- ☐ PURGE CONNECTION SIZE & TYPE
- ☐ PURGE m³/h REQUIRED
- ○ OIL MIST COOLING ○ GAS COOLING
- ○ CONTRACT GUARD TO BE USED DURING SHOP TEST (E.4.5)
- ○ ADDITIONAL GUARD DETAILS (2.3.5)

ACCESSORIES

- ○ PRIME EQUIPMENT SUPPLIER TO FURNISH ONE SET OF PLUG AND RING GAGES (APPENDIX D)
- ○ DRILL TEMPLATE FOR INTEGRAL FLANGED HUBS BY (2.1.5)
  - ○ COUPLING MANUFACTURER ○ PURCHASER
- ○ TWO-PIECE STOP RINGS BY COUPLING MANUFACTURER (3.3)
- ● HYDRAULIC INSTALLATION/REMOVAL TOOLING (TO INCLUDE HAND PUMP(S), PRESSURE GAUGE(S), FITTINGS AND HOSE(S)) BY (3.1)
  - ○ PURCHASER ○ COUPLING MANUFACTURER
- ○ PULLER BY COUPLING MANUFACTURER (3.4)
- ● MOMENT SIMULATOR (2.1.7.1) FOR TURBINE DRIVER

APPLICABLE SPECIFICATIONS

● API-671, SPECIAL PURPOSE COUPLINGS
○

(1) DUPLICATE TO TRAIN - G

○ COORDINATION MEETING ATTENDANCE REQ'D (5.2.2)

PREPARATION FOR SHIPMENT

- ● OUTDOOR STORAGE FOR MORE THAN 6 MONTHS (4.5.2)
- ● EXPECTED STORAGE TIME
- SHIPPING: (4.5.3) ○ DOMESTIC ● EXPORT
- STORAGE: (4.5.2) ○ INDOOR ● OUTDOOR
- ○ SEE COUPLING PURCHASE ORDER FOR PRESERVATION, BOXING, AND SHIPPING INSTRUCTIONS